Structured Linear System

The linear system of equations is described by

- $2x_k - x_{k+1} = y_k$, for $k = 1$
- $-(1/2)x_{k-1} + 2x_k - x_{k+1} = y_k$, for $k = 2, 3, \ldots, n-1$
- $-(1/2)x_{k-1} + 2x_k$ for $k = n$.

Problem

Formulate this linear system into $Ax = y$. Write computer code to solve the system when $y \in \mathbf{R}^n$ is given.

Implementation

Goal: Fill in the following function then submit your function in grader:

```
import numpy as np

def solve_structured_lineq(y, n):
  ...
  return A, x
```

Constraints

- y is in range of [-5,5], $10 \leq n \leq 2000$